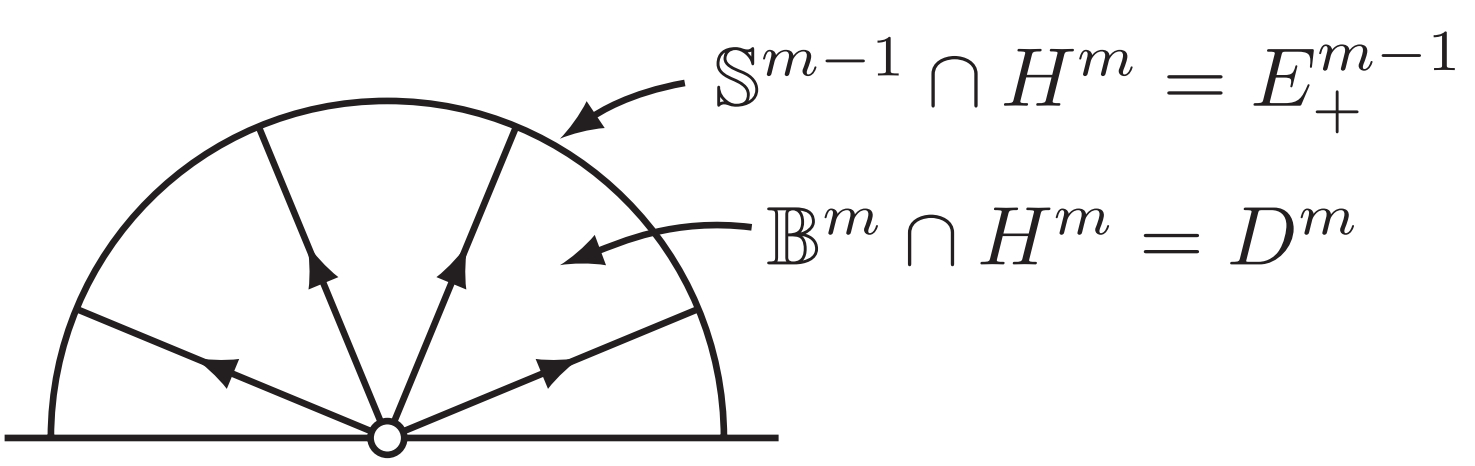

$\mathbb{S}^{m-1} \cap H^m = E_+^{m-1}$
$\mathbb{B}^m \cap H^m = D^m$